

G01 / ASSESSMENT

G01 GATE A

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

YOUR STANDALONE GATE BOOK

Independent data + Excel checkpoint

Assessment rule: Save the current attempt before opening its review. Fresh Gate B stays closed until explicitly assigned after repair.

What this checkpoint decides

This checks the foundations taught in M00-M01. It is not a job-readiness certificate. Attempt it after the lesson practices and changed retries. Keep the worked solutions closed while making your first attempt. Reading the question and using your ready-made revision notes is allowed; copying the checkpoint review and presenting it as independent work is not.

Create a fresh workbook and preserve the source below on a sheet called GateSource. Keep a separate working sheet. You may use desktop Excel's formula help for syntax, but record any hints or assistance you used. There is no speed requirement. Save the original attempt even if you later discover an error.

The source and business rules

The fictional shop supplies this delivery. One row represents one order containing one product kind. IDs identify orders and must stay text. Quantity must be a positive whole number; unit_price must be numeric and nonnegative. All amounts share one currency. Ignore taxes, discounts and delivery. An empty payment status means unknown and is allowed.

order_id	item	quantity	unit_price	ordered_on	is_paid
E001	Notebook	2	45.00	2026-09-12	TRUE
E002	Pen	4	9.00	2026-09-12	FALSE
E003	Folder	3	20.00	2026-09-12	(missing)
E004	Marker	2	25.00	2026-09-12	TRUE
E005	Pencil	5	6.00	2026-09-12	FALSE

The approved category mapping is Notebook/Paper, Folder/Paper, Pen/Writing, Marker/Writing and Pencil/Writing. Each item has exactly one supplied category. For a separate cleaning demonstration, use the original label " Pen " with one ordinary space on each side; preserve it beside a cleaned version. Do not introduce those spaces into GateSource unless you explicitly label that separate test.

Your six deliverables

1. Source and meaning. Save GateSource without overwriting it. State the grain, intended key, field types, record count, missing ID count and repeated-key result. Explain why the header is not an order and why missing status must stay distinct from FALSE.

2. Quality and cleaning. State the quantity/price rules, inspect every record and report the unknown status. Show original " Pen " and its permitted trimmed result. Explain why correcting an unknown payment by guessing would be different from trimming permitted spaces.
3. Calculation. Produce reference-based amount and payment_group columns. Report the whole amount, recorded paid, recorded unpaid, unknown and known-status amounts. Include counts by payment group. Inspect the first and last formulas. Change one quantity temporarily, predict the effect, verify it and restore the source value.
4. Summary and visual. Use an exact category lookup with a fixed mapping range. Check the mapping key. Build a native PivotTable summing amount by category and a labelled bar or column chart of those categories. Keep the grand total out of the chart's category list. Explain what the chart measures and one conclusion it cannot support.
5. Validation. Reconcile source and output counts, category counts, payment amounts, category amounts and full total. Calculate at least one row independently. Explain which check would catch a dropped row and which would catch unknown misclassified as unpaid.
6. Handover. Save and reopen the XLSX. Write a short explanation for the owner containing the result, source date, unknown order and limits. Record your evidence path and assistance used in a saved text file. Keep a backup of this record and the actual workbook.

Evidence bundle

Use an evidence folder named G01-A-attempt-01 containing gate-work.xlsx and handover.txt. The workbook needs GateSource, working calculations, mapping, PivotTable and chart. The text file needs your rule statement, checks, observed results, unknown issue and explanation. Exact sheet names other than GateSource may differ if the relationships are clear. A screenshot can supplement the files; it does not replace inspectable formulas and source data.

How to assess an honest attempt

For each of the six dimensions below, choose the best-supported level. Level 0: no attempt or no usable evidence. Level 1: followed the review but cannot yet reproduce the reasoning. Level 2: partly independent but still needs a material hint or has an unresolved error. Level 3: independently correct, inspectable and explainable, including a changed check. Level 4: meets level 3 and clearly explains the limits and detects a deliberately introduced mistake.

Dimension	Evidence needed for level 3
Source, grain, key, preservation	unchanged source, correct row meaning, ID checks and type explanation
Quality and safe standardisation	stated rules, justified trim, separate unknown issue, no invented payment
Calculation and inclusion logic	correct formulas and group logic, including known status and input-change test

Summary, lookup, PivotTable and chart

exact valid mapping, correct native pivot, suitable labelled chart and interpretation

Independent validation	source comparisons and reconciliations with stated limits
Evidence and explanation	saved/reopened files, readable handover, retained attempt and assistance record

Do not average these levels. Ready to proceed requires all six at level 3 or above and no unresolved critical failure. Critical failures include damaged source, wrong grain, an unreconciled result, invented evidence or presenting an unknown status as a confirmed payment fact. If software prevented the native pivot task, record that dimension as pending rather than awarding an unseen result.

Your recorded levels are self-assessed. Filling in the rubric does not provide independent inspection of your workbook or certify a hiring outcome. If another person reviews your files, record their feedback separately. Their name is not required for private self-study; do not fabricate a reviewer.

Repair without erasing the first attempt

Save the current attempt. Open the explained review only now. Identify the specific failed dimension, revisit its lesson and append a repair record. Then complete the changed micro-task below for that skill. A corrected copy is useful, but it is not a fresh independent first attempt. Full new assessments B and C are reserved for a deliberately assigned later reassessment; they are not opened automatically by a timer or low score.

Changed micro-task after repair

Use three new orders: F001 Pen 2 at 11 TRUE; F002 Folder 1 at 24 FALSE; F003 Notebook 2 at 38 missing. All dates are 2026-09-13. The owner asks for known-status amount, including TRUE and FALSE, plus unknown separately. Preserve source, calculate amounts, produce Paper/Writing totals using the same mapping, reconcile and explain the different inclusion question. Keep this as a new attempt under the same checkpoint history.

Progression note

Completing G01 establishes a small foundation for SQL learning. M02-M19 remain on the standalone roadmap and require their own teaching, software and assessment work. This batch does not mark those later stages complete or present legacy ticks as evidence for the rebuilt lessons.